

Lab 1 Safety and Setup

(This lab is for safety and setup. It will not be assessed. To avoid affecting the following labs, please complete the assembly steps before next lab.)

1. Introduction

This lab introduces the unmanned aerial vehicle (UAV) platform used in CDE1302. You will complete the required safety training, identify the main components, assemble the UAV.

Before conducting any UAV programming or flight test, students must understand the following safety instructions:

1. The hazards associated with batteries, propellers, motors, and autonomous flight.
2. The correct procedures for responding to abnormal or emergency situations.
3. The purpose and function of the main UAV components.
4. How the UAV components are mechanically and electrically integrated.
5. How to inspect and assemble the UAV safely.

No free flight should be performed during this lab unless specifically authorised and supervised by the teaching staff. Please bring your own computer for this lab.

2. Tasks

In this lab, you need to complete following tasks.

1. Safety briefing.
2. UAV System Briefing
3. Form groups, collect and inspect drone hardware set.
4. Drone hardware assembly.

3. General Safety Briefing and Safety Requirements

Students must follow all instructions given by the lecturer, teaching assistant, and laboratory staff.

3.1 Personal Safety

1. Tie back long hair and secure loose clothing.

2. Remove loose jewellery before working near motors and propellers.
3. Keep fingers, tools, cables, and other objects away from the propeller area.
4. Do not touch rotating motors or propellers.
5. Do not work alone when conducting powered UAV tests.
6. Do not operate the UAV when feeling unwell or unable to concentrate.

3.2 Propeller Safety

Propellers rotate at high speed and can cause injury or damage.

Students must:

1. Remove the propellers during configuration, programming, calibration, and communication tests whenever propellers are not required.
2. Install propellers only when instructed by the teaching staff.
3. Check each propeller for cracks, deformation, or looseness.
4. Confirm that each propeller is installed with the correct orientation on the correct motor.
5. Install the prop guards before any authorised flight test.
6. When the UAV is powered, the paired RC must have the kill switch ON outside of flight tests.
7. Keep everyone outside the designated flight area while the UAV is armed.

3.3 Battery Safety

Lithium-polymer batteries can cause fire, burns, or equipment damage if they are misused.

Students must:

1. Inspect the battery before use.
2. Do not use a battery that is swollen, punctured, leaking, excessively hot, or physically damaged.
3. Do not short-circuit the battery terminals.
4. Connect the battery only after receiving permission from the teaching staff.
5. Use only the approved charger and charging settings.
6. Charge the battery in the designated charging area.
7. Do not leave a charging battery unattended.
8. Disconnect the battery immediately after completing the test.
9. Report abnormal battery temperature, smell, smoke, or swelling immediately.
10. Place damaged or suspicious batteries in the designated safe area. Do not place them in a normal rubbish bin.

3.4 Arming and Disarming

A UAV is considered armed when the propulsion system is enabled and the motors may rotate.

Students must understand that:

1. An armed UAV must always be treated as active.
2. The UAV must not be armed on a desk or crowded workbench.
3. The operator must announce clearly and count down before arming or starting mission
 - “**Arming** in 3...2...1”
 - “**Starting mission** in 3...2...1”
4. Everyone must move outside the safety area before arming.
5. The operator must be ready to disarm or activate the kill switch immediately.
6. **The UAV should be Disarmed after landing or whenever the test is stopped.**
It is strongly recommended to **Activate** the RC kill switch immediately afterward as an additional safety measure.
7. The RC kill switch must be **Activated** outside of flight tests.

3.5 Laboratory Safety

1. Keep the work area organised and free of unnecessary items.
2. Do not conduct flight tests outside the designated flight area.
3. Do not enter the flight area without permission.
4. Report damaged equipment immediately.
5. Do not modify hardware or software parameters without approval.
6. Stop the test immediately if unexpected behaviour is observed.

3.6 Flight Test Roles

Assign the following roles before any flight test.

Role	Responsibility
Pilot or safety operator	Controls the remote controller and disarms during an emergency
Ground-control operator	Monitors the UAV status and software
Observer	Watches the surrounding area and identifies hazards
Recorder	Records observations, warnings, and test results

3.7 Safety Zone

Before any flight test, students must be clear on the following:

- UAV take-off point, mission profile, landing point
- Flight boundary
- Team members location and readiness
- Emergency exit
- Fire extinguisher or battery safety equipment

3.8 Emergency Situations

Complete the table based on the safety briefing.

Q1. Complete the table based on the safety briefing	
Situation	Immediate Response
UAV moves unexpectedly before take-off	
UAV tips over during motor operation	
UAV flies toward a person	
Communication with the ground station is lost	
Remote-control connection is lost	
Position estimation becomes unstable	
A propeller becomes loose	
A motor produces abnormal sound	
The battery becomes hot or swollen	
Smoke is observed	
UAV leaves the designated flight area	

4. UAV System Briefing

4.1 Basic UAV System Architecture

The UAV system contains the following subsystems:

- Mechanical structure
- Propulsion system
- Power system
- Flight-control system
- Sensor system
- Communication system
- Ground control system

The UAV System Architecture illustrated as follows:

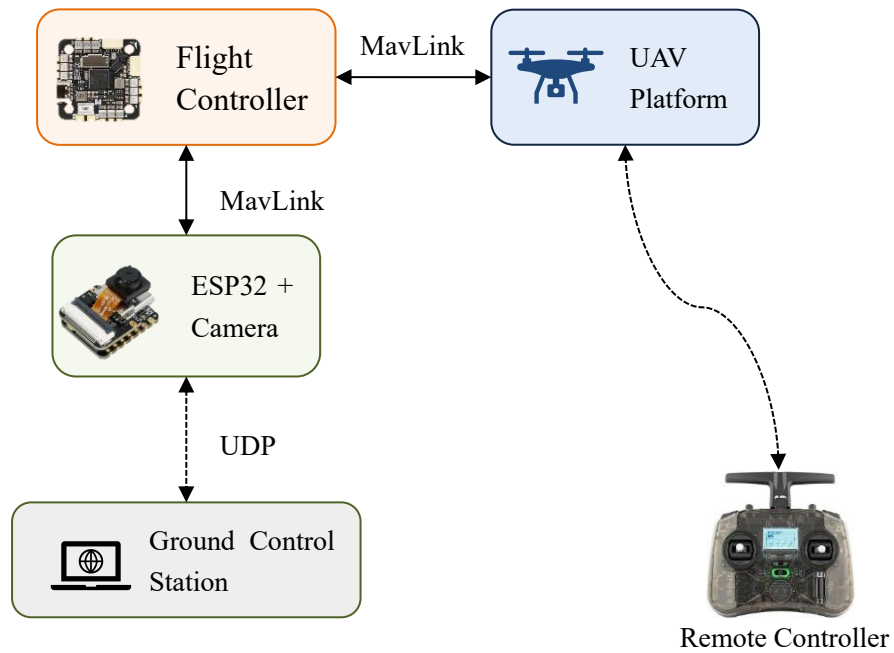


Figure 4.1 UAV System Architecture

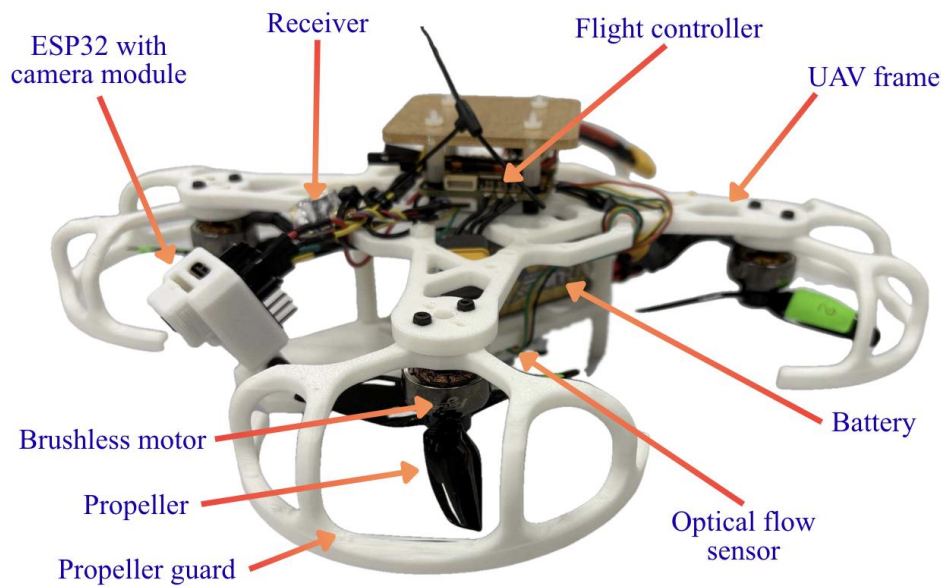


Figure 4.2 An example of the assembled UAV system

4.2 Flight Controller and Avionics

The flight controller receives information from onboard sensors, estimates the UAV state, calculates the required control commands, and sends output signals to the motors through the electronic speed controllers. The flight controller used is MicoAir743v2-AIO-45A, which is a high-performance H743 FC and 32bit ESC All-In-One supports multiple firmware: Ardupilot/**PX4**/INAV/Betaflight



Figure 4.3 MicoAir743v2-AIO-45A

Table 4.1 Key specifications of the Flight Controller (MicoAir743v2-AIO-45A)

Item	Description
MCU Chip	TM32H743VIH6, 480MHz, 2MB Flash
IMU Sensor	BMI088+BMI270
Baro Sensor	SPL06
Compass	No onboard compass, external connection is supported
Input Voltage	2S – 6S Li-Po Battery
UART	7 full-function serial ports (UART1, UART2, UART3, UART4, UART5, UART6, UART8)
USB	Type-C
Size	36 × 36 × 8mm
Weight	10g

Manufacturer link: https://micoair.com/flightcontroller_micoair743v2_aio_45a/

4.3 Sensors

4.3.1 Optical-Flow and Range Sensor



Figure 4.4 MTF-02p

2IN1 sensor, small size with high performance, available for outdoor use

Explain that an optical-flow sensor:

- Observes the ground surface.
- Detects apparent image movement.
- Helps estimate horizontal motion.
- Can support position control when GPS is unavailable.

Important limitations:

- It needs sufficient lighting.
- It needs visible ground texture.
- It may perform poorly over reflective, transparent, dark, or uniform surfaces.
- Its lens must not be blocked.

Manufacturer link: https://micoair.com/optical_range_sensor_mtf-02p/

4.3.2 Camera Detection Sensor

The camera is used for:

- Image collection
- Object detection
- AprilTag detection
- Perception-guided missions

Clarify that a camera does not automatically provide useful control information. Images must be processed by a computer or onboard processor before they can be used for navigation.

4.4 Power Subsystem

Lithium-Polymer Battery:

- The battery stores electrical energy.

- Battery voltage determines compatibility with the propulsion and electronic systems.
- Battery capacity affects possible operating time.
- Actual flight time depends on UAV weight, motor demand, battery condition, and flight behaviour.

Battery markings:

- Number of cells
- Nominal voltage
- Capacity in milliampere-hours
- Discharge rating
- Connector type

Battery Safety:

- Do not use swollen, punctured, leaking, or unusually hot batteries.
- Avoid short circuits.
- Use only an approved charger.
- Do not leave charging batteries unattended.
- Disconnect the battery after testing.

Examples of:

- A healthy battery
- A damaged connector
- A swollen battery, using a photograph or retired training battery rather than an unsafe active battery

4.5 Communication and Ground Control Subsystem

The ground control station allows the operator to monitor the UAV, configure parameters, view sensor information, upload missions, and issue authorised commands. We will make use of the ground control station in the next lab.

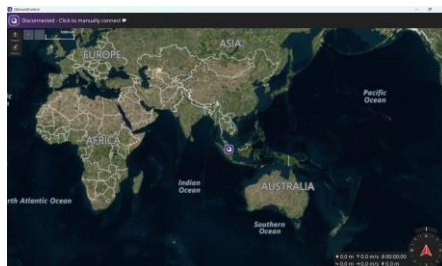


Figure 4.5 GUI of Ground Control Station

5. Lab Task 1: Grouping and Hardware Inspection

1. Form a group of 3-4 students (both students should come from the same lab session) and sign up your group on the course Canvas page.
2. Collect your hardware set from the lab and check the list.
3. Inspection and identification of parts for hardware set.

Each group will be provided with the following items as shown in Figure 5.1:

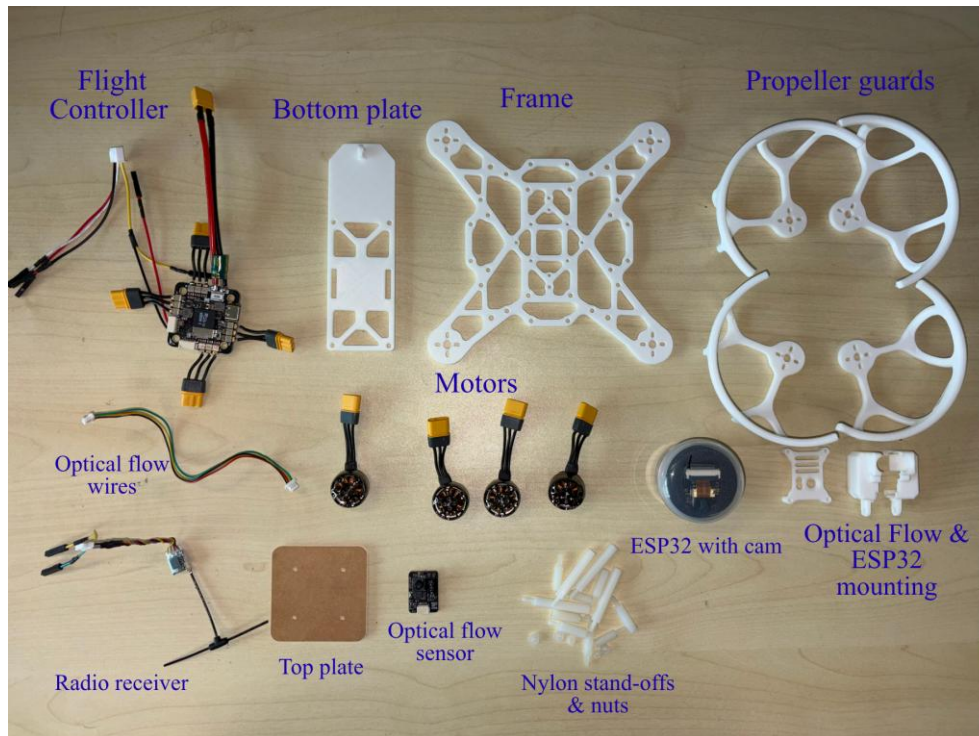


Figure 5.1 Illustration of components

The detailed component list is given below:

Table 5.1 Component list

No.	Item	Quantity
1	Base drone frame	1
2	Prop guards	4
3	Battery and camera mount	1
4	Optical flow sensor mount	1
5	ESP32-S3 Sense cover (left and right)	1
6	ESC + FC acrylic cover	1
7	ESC + FC board	1
8	Motors	4
9	Optical flow sensor	1

10	ESP32-S3 Sense	1
11	ELRS receiver	1
12	M2x30 nylon stand-off	4
13	M2.5x15 nylon stand-off	4
14	M2.5x30 nylon stand-off	4
15	M2 nylon nut	4
16	M2.5 nylon nut	4
17	M2x10 screw	12
18	M2x15 screw	4
19	M3x15 screw	1
20	M3 flat washer	3
21	M3 spring washer	1
22	M3 nut	1

Check that you have all the parts listed in the table above.

6. Lab Task 2: Component Assembly

Carry out the assembly under the supervision of the teaching assistant.

6.1 Assembly Precautions

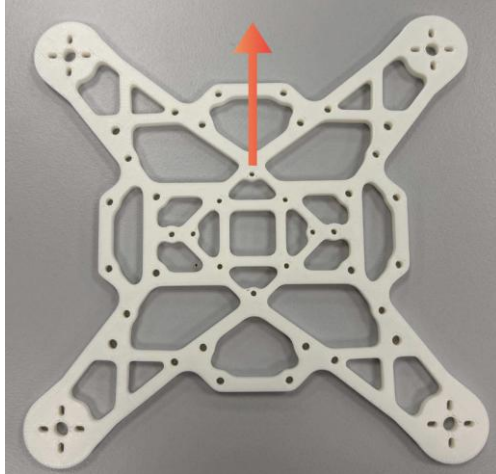
Before starting:

1. Ensure that the battery is disconnected.
2. Remove all propellers.
3. Place the UAV on a stable work surface.
4. Do not overtighten screws.
5. Do not force connectors into sockets.
6. Check connector orientation before insertion.
7. Avoid pulling directly on wires.
8. Use cable ties only where required.
9. Ensure that cables cannot enter the propeller area.

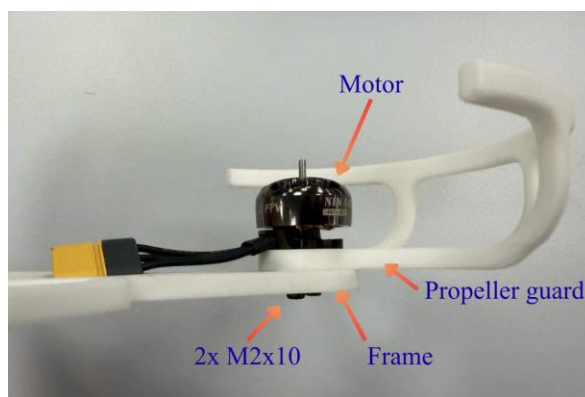
6.2 Frame and Motor Assembly

Complete the following tasks as instructed:

1. Identify the front direction of the UAV.



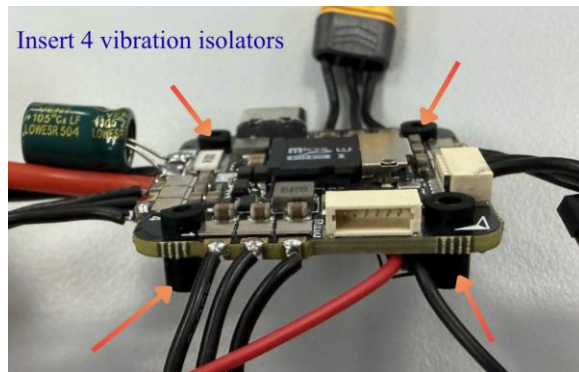
2. Identify the motor positions.
3. Confirm the motor numbering convention.
4. Check the required motor rotation direction.
5. Secure the motors to the frame with prop guards, where the guard is between the frame and the motor, using M2x10 screws.



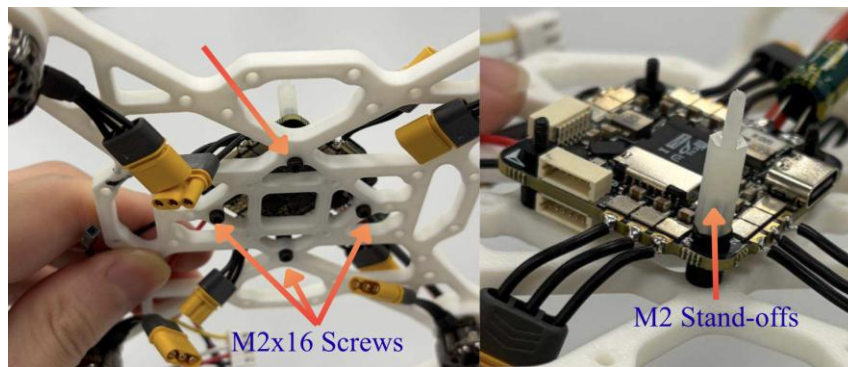
6. Check that the motor cables are correctly routed.
7. Verify that no cable is trapped or excessively stretched.

6.3 Flight Controller Installation

1. Insert four vibration isolators into four holes on the flight controller.



2. Mount the flight controller on the frame (opposite side of the motors) using M2x15 screws and M2 stand-offs. The screws go through the frame and isolator from the bottom side and are tightened with the stand-offs. Make sure the front direction of the controller (labeled on the board) aligns with the frame's front direction.



Check the following:

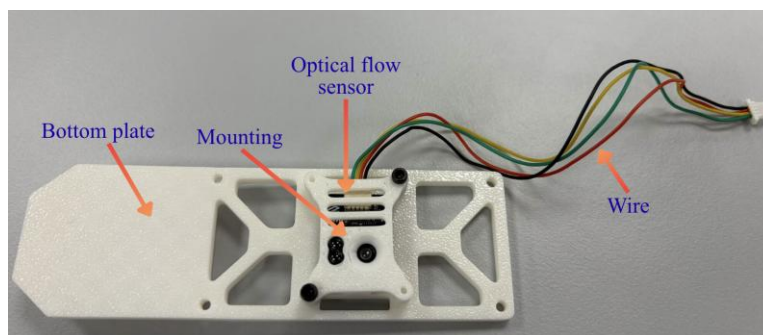
1. The flight controller is installed in the correct orientation.
2. The forward arrow points toward the UAV front.
3. The flight controller is mounted securely.
4. Vibration-isolation materials are installed correctly and not over tightened.
5. Cables do not place excessive force on the controller.
6. The USB port and other required connectors remain accessible.

Check	Pass	Fail	Remarks
Correct controller orientation	☐	☐	
Controller firmly mounted	☐	☐	

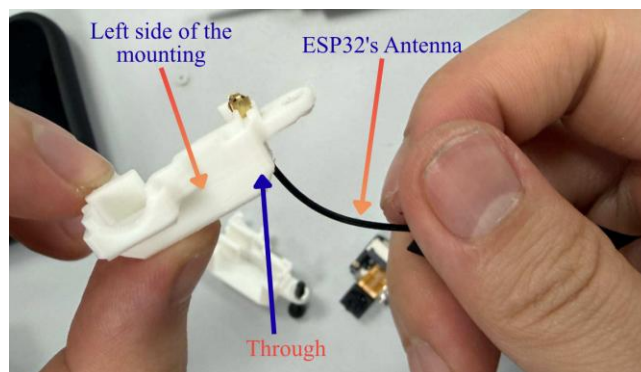
Vibration isolation installed	☐	☐	
Cables safely routed	☐	☐	
Connectors remain accessible	☐	☐	

6.4 Sensor Installation

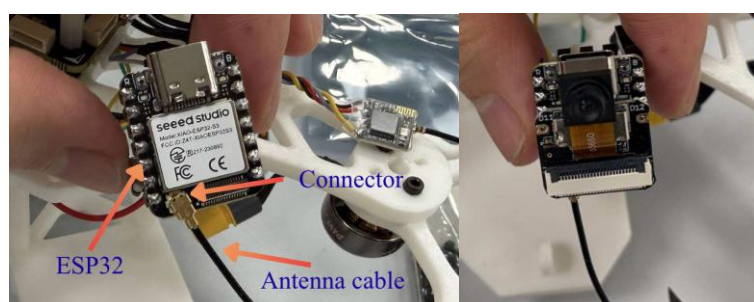
1. Connect an optical flow wire (the middle wire in its box) to the sensor. Install the optical flow sensor with the bottom plate and its mounting as shown in the figure using four M2x10 screws.



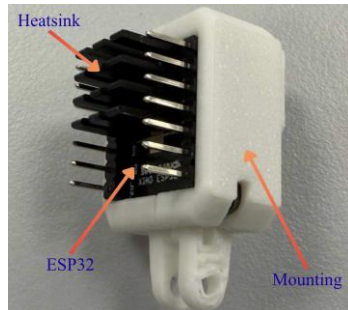
2. ESP32:
 - a. Put the antenna cable through the left side of its mounting.



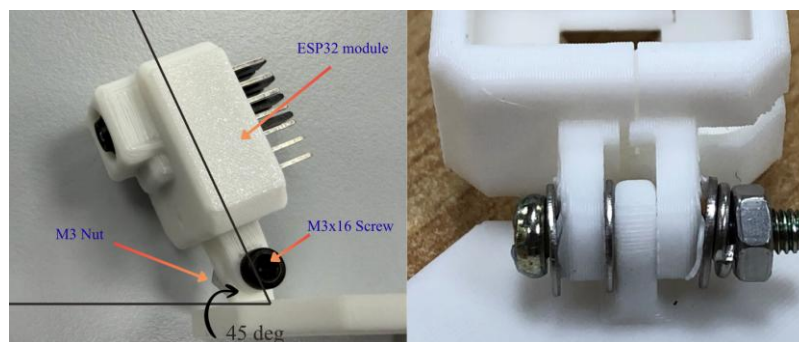
- b. Connect the antenna cable to the ESP32 then put the camera module onto it.



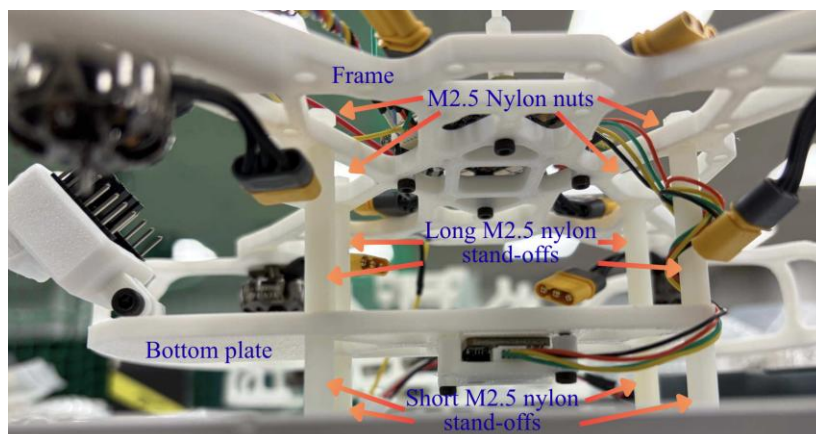
- c. Put the ESP32 and its mounting together. Attach the heatsink included with the ESP32 inside the case to the back of the ESP32 as in the following picture.



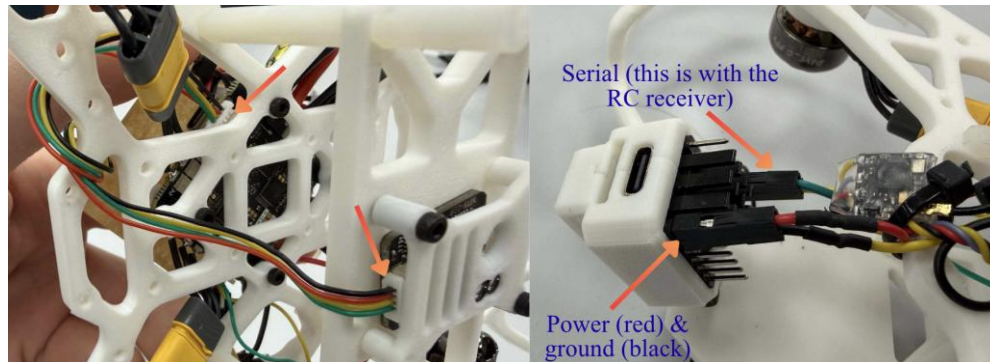
- d. Mount the ESP32 module with the bottom plate using an M3x15 screw and an M3 nut. Tilt the ESP32 module as in the right figure. Tip: Insert washers in the middle to make the mounting more rigid (left figure).



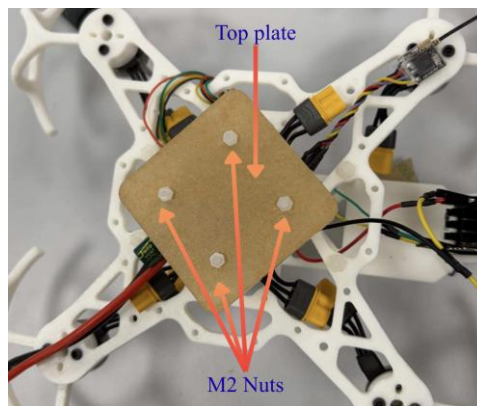
3. Mount the bottom plate to the frame with four long M2.5 nylon stand-offs and M2.5 nuts as shown in the figure below. The short M2.5 nylon stand-offs are used as landing gear. Note that the ESP32 module should be on the front side of the frame.



4. The ESP32's antenna has 2-sided tape. Attach the antenna to a convenient location.
5. Connect the cables.



6. Install the top plate on the four M2 stand-offs we installed earlier on the flight controller and tighten it using four M2 nuts.



7. Ensure no cable is in the propeller operational space at any time. Use cable ties to secure it.

Inspect or install the sensors as instructed.

Sensor	Correct Orientation	Securely Mounted	Lens or Surface Clean	Cable Connected
Optical-flow and range sensor	☐	☐	☐	☐
Camera	☐	☐	☐	☐
Other: _____	☐	☐	☐	☐

Explain why the orientation and mounting position of the optical-flow sensor are important.

6.5 Communication Module (ESP32)

Identify the communication link used by the UAV.

Communication module: _____

Communication method:

☐ Wi-Fi

☐ Telemetry radio

☐ USB

☐ Other: _____

Check that:

- The communication module is securely mounted.
- The cable is connected to the correct flight-controller port.
- The antenna is not blocked by conductive materials.
- The antenna is kept away from propellers.
- The cable does not interfere with sensors.

6.6 Prop Guard Installation

Inspect the installed prop guards using the checklist below.

Inspection Item	Pass	Fail	Remarks
All prop guards are installed	☐	☐	
Prop guards are firmly secured	☐	☐	
Prop guards do not touch motors	☐	☐	

Prop guards do not block sensors	☐	☐	
Prop guards do not restrict propeller rotation	☐	☐	

7. Lab Task 3: Electrical Connection Inspection

Do not connect the battery until the teaching assistant has checked the UAV.

7.1 Connection Checklist

Connection	Checked	Remarks
Battery connector	☐	
Flight-controller power connection	☐	
Motor and ESC connections	☐	
Receiver connection	☐	
Optical-flow and range sensor connection	☐	
ESP32 connection	☐	

7.2 Wiring Inspection

Confirm that:

- No bare conductor is visible.
- No wire is damaged or excessively bent.
- Connectors are fully inserted.
- Polarity-sensitive connectors are correctly oriented.
- Wires are secured away from propellers.
- Signal and power cables are connected to the intended ports.

8. Final Assembly Inspection and GA Verification

Before leaving the laboratory, each group must ask the Graduate Assistant (GA) to inspect the completed UAV assembly.

If the assembly cannot be completed during Lab 1, the group must complete all remaining assembly work and ensure that the UAV is fully assembled before attending Lab 2. The completed assembly should be checked by the GA before proceeding with Lab 2 activities.

The GA will check that:

1. All required components are installed correctly and securely.
2. The flight controller is installed in the correct orientation.
3. The motors and prop guards are securely mounted.
4. The optical-flow sensor and camera are installed in the correct orientation.
5. All electrical connectors are fully inserted and connected to the correct ports.
6. Cables are properly routed and kept away from the propeller sweep area.
7. No loose screws, nuts, stand-offs, or other components are present.
8. No visible damage is present on the UAV, connectors, wires, or sensors.
9. The battery remains disconnected during the inspection.
10. The UAV is safe and ready for the next laboratory session.

Any issue identified by the GA must be corrected before the group proceeds to the subsequent laboratory session.

Final Check	Pass	Fail	Remarks
Mechanical assembly	☐	☐	
Flight-controller orientation and mounting	☐	☐	
Sensor installation	☐	☐	
Electrical connections	☐	☐	
Cable routing	☐	☐	
Prop guards	☐	☐	
Overall assembly	☐	☐	

GA Verification

GA Name: _____

Signature: _____

Date: _____